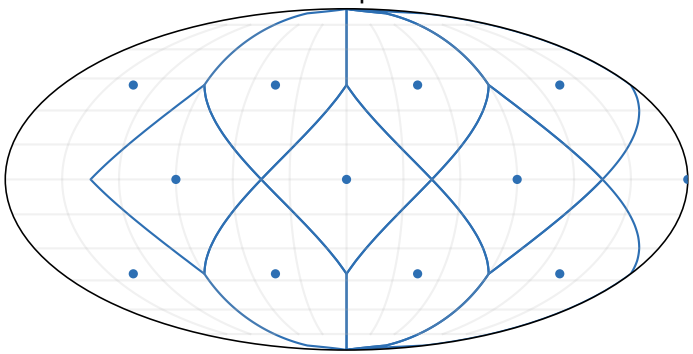
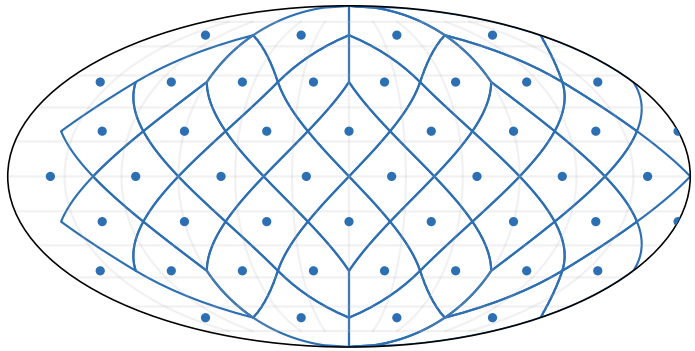
Equivolumetric partition of SO(3): equal-area Hopf–HEALPix base vs. naïve equal-angle grid

OURS — equal Haar-volume cells (max/min volume ratio = 1.000, exact by construction)

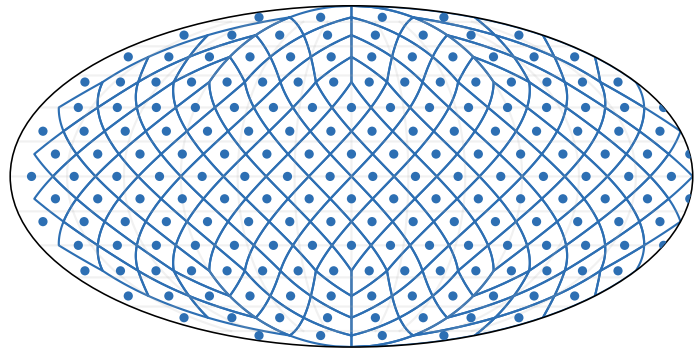
level 0 · nside 1 · 12 S² pixels × 6 roll = 72 cells



level 1 · nside 2 · 48 × 12 = 576 cells

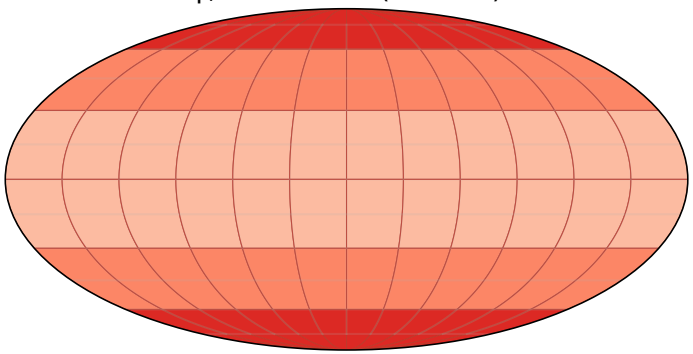


level 2 · nside 4 · 192 × 24 = 4608 cells

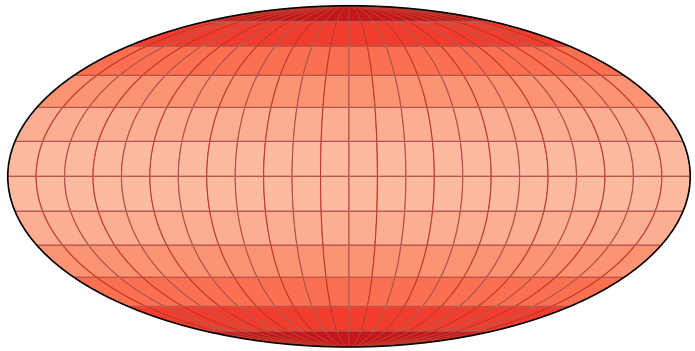


NAÏVE equal-angle (φ,θ) grid — cells pinch at the poles (dark = tiny-volume cell); uniform categorical ≠ uniform density

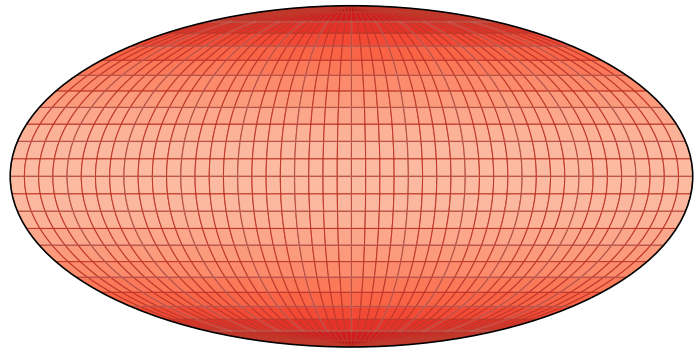
φ,θ = 12 × 6 (72 cells)



24 × 12 (288 cells)



48 × 24 (1152 cells)



Consequence (SO(3), Haar measure): naïve grid max/min cell-volume ratio ≈ 3.7× → 7.6× → 15.3× and grows without bound as θ-resolution rises. A uniform categorical over naïve cells then carries a +0.11–0.14 nat log-likelihood bias vs. the true uniform density; our equal-volume tree has bias 0 (exact). For scale, the SYMSOL-I SOTA margin is ≈0.4 nat — so the discretization bias is not negligible at the precision we compete at.